Experiment No : 1

Title: Perform following String operations: a. substring b. palindrome c. compare d. copy e. Reverse.

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Program:

```
#include <stdio.h>
#include <string.h>
#include <stdlib.h>

/* Function Prototypes */
void enterstring(char input[]);
int palindrome(char input[]);
void copy(char input[] , char output[]);
int stringlength(char input[]);
void substring(char input[] , char output[]);
void reverse(char input[] , char output[]);

int main()
{
   int choice, c, l ;
   char str[100] , copied[100] , ss[100] , rev[100];
   //Initialize();
  do
   {
      printf("\n\t\t\t# STRING OPERATIONS #             ");
      printf("\n----------------------------------------");
      printf("\n1. Insert a string");
      printf("\n2. Check if the string is Palindrome");
      printf("\n3. Copy the string");
      printf("\n4. Calculate length of the string");
      printf("\n5. Create a substring of the string");
      printf("\n6. Reverse the string");
      printf("\n7. Exit");
      printf("\n\nEnter your choice : ");
      scanf("%d", &choice);
```

```
switch(choice)
{
    case 1:
        enterstring(str);
        printf("The entered string: %s\n", str);
        break;

    case 2:
        c = palindrome(str);
        if (c==1)
          printf("The string is palindrome \n");
        else
          printf("The string is not palindrome \n");
        break;

    case 3:
        copy(str ,copied);
        printf("The copied string is : %s \n" , copied);
        break;

    case 4:
        l = stringlength(str);
        printf("Length of the string is : %d \n" , l);
        //printf("Length of the string is: %d\n", stringlength(str));
        break;

    case 5:
        substring(str , ss);
        printf("The substring is : %s \n", ss);
        break;

    case 6:
        reverse(str ,rev);
        printf("The reversed string is : %s \n" , rev);
        break;


    case 7:
        return 0;
```

```
            default:
                printf("\nInvalid Choice!");
        }
    }while(1);
return 0;

}

void enterstring(char input[])
{
    printf("Enter the string: ");
    scanf("%99s", input);

}

int palindrome(char input[])
{
    int i= 0, j= 0;
    while (input[j]!='\0')
       j++;
    j--;
    while(i<j)
    {
      if(input[i]!= input[j])
        return 0;
      i++;
      j--;
    }

return 1;
}

void copy(char input[] , char output[])
{
    int i ;
    for(i = 0 ; input[i] != '\0' ; i++)
    {
        output[i] = input[i];
    }
```

```
    output[i] = '\0';
}

int stringlength(char input[])
{
    int i = 0;
  while(input[i] != '\0')
  {
      i++;
  }
  return i;
}

void substring(char input[] , char output[])
{
    int n, i = 0, p;
    printf("--NOTE--\nThe indices of the string start from 0\n");
    printf("\nEnter the index from where the substring is to be created : ");
    scanf("%d", &p);
    n = stringlength(input);
    if (p >= n)
    {
        printf("Invalid index!\n");
        return;
    }
    while (input[p+i] != '\0')
    {
        output[i] = input[p+i];
        i++;
    }
    output[i] = '\0';
}

void reverse(char input[] , char output[])
{
    int i , n = 0 ;
    n = stringlength(input);
    for(i = 0 ; i<n ; i++)
    {
        output[i] = input[n - 1 - i];
```

```
    }
    output[i] = '\0';
}
```

Output :

```
           # STRING OPERATIONS #
----------------------------------------
1. Insert a string
2. Check if the string is Palindrome
3. Copy the string
4. Calculate length of the string
5. Create a substring of the string
6. Reverse the string
7. Exit

Enter your choice : 1
Enter the string: madam
The entered string: madam
```

```
           # STRING OPERATIONS #
----------------------------------------
1. Insert a string
2. Check if the string is Palindrome
3. Copy the string
4. Calculate length of the string
5. Create a substring of the string
6. Reverse the string
7. Exit

Enter your choice : 2
The string is palindrome
```

```
           # STRING OPERATIONS #
----------------------------------------
1. Insert a string
2. Check if the string is Palindrome
3. Copy the string
4. Calculate length of the string
5. Create a substring of the string
6. Reverse the string
7. Exit

Enter your choice : 1
Enter the string: shamiksha
The entered string: shamiksha
```

```
           # STRING OPERATIONS #
----------------------------------------
1. Insert a string
2. Check if the string is Palindrome
3. Copy the string
4. Calculate length of the string
5. Create a substring of the string
6. Reverse the string
7. Exit

Enter your choice : 2
The string is not palindrome
```

```
          # STRING OPERATIONS #
----------------------------------------
1. Insert a string
2. Check if the string is Palindrome
3. Copy the string
4. Calculate length of the string
5. Create a substring of the string
6. Reverse the string
7. Exit

Enter your choice : 3
The copied string is : shamiksha
```

```
          # STRING OPERATIONS #
----------------------------------------
1. Insert a string
2. Check if the string is Palindrome
3. Copy the string
4. Calculate length of the string
5. Create a substring of the string
6. Reverse the string
7. Exit

Enter your choice : 4
Length of the string is : 9
```

```
            # STRING OPERATIONS #
----------------------------------------
1. Insert a string
2. Check if the string is Palindrome
3. Copy the string
4. Calculate length of the string
5. Create a substring of the string
6. Reverse the string
7. Exit

Enter your choice : 5
--NOTE--
The indices of the string start from 0

Enter the index from where the substring is to be created : 4
The substring is : iksha
```

```
          # STRING OPERATIONS #
----------------------------------------
1. Insert a string
2. Check if the string is Palindrome
3. Copy the string
4. Calculate length of the string
5. Create a substring of the string
6. Reverse the string
7. Exit

Enter your choice : 6
The reversed string is : ahskimahs
```

```
          # STRING OPERATIONS #
----------------------------------------
1. Insert a string
2. Check if the string is Palindrome
3. Copy the string
4. Calculate length of the string
5. Create a substring of the string
6. Reverse the string
7. Exit

Enter your choice : 7


=== Code Execution Successful ===
```